

Dr. Marcus T. Hayes

CHIEF MEDICAL OFFICER / VP CLINICAL
INFORMATICS

Email: marcus.hayes@healthtech.md

Phone: +1 (617) 925-4410

Location: Boston, MA 02115

LinkedIn: linkedin.com/in/dr-marcus-t-hayes

20+
Yrs Healthcare IT

M.D., Ph.D.
Credentials

\$180M
Efficiency Gains

42K
Clinicians Enabled

22
EHR Implementations

PROFESSIONAL SUMMARY

Physician-executive and clinical informatics leader with 20+ years at the intersection of medicine, technology, and health system transformation. Dual-trained MD/PhD (Harvard Medical School / MIT) combining deep clinical expertise with technical fluency in AI/ML, EHR systems, and health data engineering. Served as CMIO at Mass General Brigham, Boston Children's Hospital, and Partners HealthCare. Expert in clinical AI validation, FHIR R4 interoperability, Epic optimization, and FDA digital health regulatory pathway. Author of 45 peer-reviewed papers and 2 clinical informatics textbooks. Active practitioner (Board Certified Internal Medicine).

SKILLS

CLINICAL INFORMATICS

EHR Optimization

CDS Implementation

CPOE Design

Clinical Workflow

Physician Adoption

HEALTH DATA & INTEROP

HL7 FHIR R4

SMART on FHIR

HL7 v2

CDA/CCDA

OMOP CDM

i2b2

EHR PLATFORMS

Epic (Certified)

Cerner (Proficient)

Allscripts

Meditech

Healtheon

CLINICAL AI & ANALYTICS

Python

R

SAS

SQL

REDCap

Tableau

Power BI

DataRobot

CLOUD - HEALTHCARE

AWS HealthLake

Azure Health API

Google Healthcare API

Databricks

REGULATORY

FDA 510(k)/DeNovo

HIPAA/HITECH

ONC 21st Century Cures

CMMC Healthcare

CERTIFICATIONS

Board Certified - Internal
Medicine (ABIM, 2006)

Board Certified - Clinical
Informatics (ABPM/ABP, 2009)

Epic Certified Physician
Builder (2015)

CPHIMS - HIMSS (2012)

CHCIO - CHIME (2018)

AWS Healthcare Competency

WORK EXPERIENCE

Chief Medical Officer / VP Clinical Informatics Mass General Brigham · Boston, MA

Jul 2019 -

Present

- Serve as CMO and VP Clinical Informatics for \$18B integrated health system with 78,000 employees across 16 hospitals and 450+ outpatient sites
- Led \$380M Epic EHR optimization program (2020-2023): reduced physician documentation burden by 38% (1.8M hours recovered annually), improved clinical decision support accuracy to 96.2%
- Deployed 14 FDA-cleared clinical AI tools across radiology, pathology, and critical care; sepsis prediction model reduced ICU mortality by 22% in prospective validation study
- Built MGB's enterprise FHIR R4 data platform integrating 180+ external health systems; enabled real-time care coordination for 4.2M attributed patients
- Established MGB's AI Clinical Validation and Oversight program — first academic medical center to publish prospective AI performance data on all deployed models
- Led COVID-19 clinical informatics response: deployed predictive surge model, remote monitoring platform, and vaccine distribution logistics system serving 2.8M patients
- Mentored 28 physician informaticists; 12 placed in CMO/CMIO roles at other health systems

Epic

HL7 FHIR

Python

R

AWS HealthLake

Azure Health

SQL

REDCap

Tableau

Chief Medical Information Officer (CMIO) Boston Children's Hospital · Boston, MA

Apr 2014 - Jun

2019

- Served as CMIO for #1-ranked pediatric hospital (US News); led 35-person clinical informatics team supporting 7,000 clinicians
- Directed \$180M Epic go-live covering all 7 BCH campuses — delivered on time, 6% under budget, highest clinical staff satisfaction score (91%) in Epic's academic medical center benchmarks
- Deployed BCH's first clinical AI programs: early warning system for pediatric sepsis (presented at NEJM Journal Watch, 2018), automated radiology triage for fracture detection
- Built pediatric-specific clinical decision support content library (4,200+ rules) — reduced potentially harmful drug dosing errors by 87% for weight-based pediatric medications

Epic

Cerner

Healtheon

Python

R

REDCap

i2b2

OMOP CDM

Medical Director, Clinical Analytics Partners HealthCare · Boston, MA

Jul 2008 - Mar

2014

- Built Partners' first enterprise clinical analytics program; designed population health dashboards used by 60+ clinical programs serving 3.5M patients
- Led meaningful use (Stage 1, 2, 3) attestation program across all Partners hospitals — secured \$280M in federal incentive payments
- Published landmark study on EHR-based early warning systems (NEJM, 2011, 1,400+ citations)

Eclipsys

Allscripts

SAS

SQL

QlikView

Epic

Resident & Fellow, Internal Medicine / Clinical Informatics Massachusetts General Hospital · Boston, MA

Jul 2003 -

Jun 2008

- Completed Internal Medicine residency and Clinical Informatics fellowship; board certified in Internal Medicine (ABIM 2006) and Clinical Informatics (ABPM/ABP 2009)

(2022)
CPHQ – NAHQ (2014)

EDUCATION

Ph.D., Biomedical Informatics
MIT – Computer Science & AI Lab
2007
AMIA Doctoral Dissertation Award
· Advisor: Prof. Peter Szolovits

M.D., Medicine
Harvard Medical School
2003
Research Distinction · AOA Honor Society

B.A., Biochemistry & Computer Science
Princeton University
1998
Summa Cum Laude · Phi Beta Kappa · Valedictorian

LANGUAGES

English	Native
Haitian Creole	Native
French	Fluent (C1)
Spanish	Conversational (B1)

AWARDS

- 🏆 HIMSS Davies Award of Excellence (2022)
- 🏆 Modern Healthcare Top 25 Clinical Informaticists (2021)
- 🏆 AMIA Distinguished Paper Award (2018)
- 🏆 NEJM Knowledge+ Editorial Board (2019–Present)
- 🏆 HMS Young Faculty Innovator Award (2016)

VOLUNTEERING

- Board of Governors – AMIA (2020–Present)
- Editorial Board – JAMIA, Journal of Med Internet Research
- Advisor – FDA Digital Health Center of Excellence
- National Academy of Medicine – Emerging Leader (2023)
- Haiti Health Innovation Foundation – Co-founder

CONTACT

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- Research focus: natural language processing of clinical notes for adverse drug event detection; PhD thesis won AMIA Doctoral Dissertation Award (2007)

KEY PROJECTS

MGB Sepsis AI System 2020–2022	Multi-model ensemble for early sepsis detection. 22% ICU mortality reduction, 94.7% AUROC. Prospective study published in New England Journal of Medicine (2023).
BCH Epic Go-Live 2016–2018	\$180M Epic implementation across all 7 BCH campuses. 7,000 clinicians, 91% satisfaction. Highest KLAS Epic academic medical center go-live score (2018).

PUBLICATIONS & TALKS

- 45 peer-reviewed papers · 8,400+ citations · h-index: 24
- 'Clinical Informatics: A Practical Guide' – Springer, 2020 (Co-author)
- NEJM: 'EHR-Based Early Warning and Intervention' (2011, 1,400 citations)
- JAMIA: 'Prospective Validation of Clinical AI at Scale' (2023, 280 citations)
- HIMSS Annual Conf Keynote – 'AI in Healthcare: A Clinician's Perspective' (2022, 2023)

PARSER VERIFICATION TABLE

Field	Value	Status
name	Dr. Marcus T. Hayes	✓ HIGH
email	marcus.hayes@healthtech.md	✓ HIGH
phone	+1 (617) 925-4410	✓ HIGH
location/address	25 Shattuck Street, Boston, MA 02115	✓ HIGH
job_title	Chief Medical Officer / VP Clinical Informatics	✓ HIGH
current_employer	Mass General Brigham	✓ HIGH
years_experience	20+	✓ HIGH
work_history_entries	4	✓ HIGH
education_entries	3	✓ HIGH
certifications_count	7	✓ HIGH
skills_groups_count	6	✓ HIGH
languages	English:Native, Haitian Creole:Native, French:Fluent (C1), Spanish:Conversational (B1)	✓ HIGH
awards_count	5	✓ HIGH
projects_count	2	✓ HIGH